

PROBLEM SET / 02

Reproducible agentic analytics for civic records

Reproducible civic analytics through an agent-neutral tool service

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Developed after the original submission and grading dates. Status: In progress.

Abstract

Natural-language civic questions often hide analytical choices: date boundaries, complaint families, geographic units, cohort selection and missing values. This implementation places those choices in a strict analytical contract. A compatible agent handles conversation, while a Python service owns validation, deterministic calculations, saved results and derived artifacts. The same operations are available through JSON CLI and seven MCP tools. [\[1\]](#) [\[2\]](#)

The service implements record filtering, grouped statistics, geographic predicates, period comparisons and background CSV exports, with Elasticsearch and Kibana adapters. Its local demonstration uses 32 authored synthetic requests. This case study reports that demonstration, historical generated-data measurements and the focused engineering verification for version 0.6.0. These evidence classes are kept separate. [\[1\]](#) [\[3\]](#) [\[4\]](#) [\[5\]](#)

Problem and design objective

The SCADS problem asks for conversational access to large civic datasets. The design objective is to make the analytical answer inspectable and reusable across agents: every reported number should resolve to a saved specification, dataset identity and result. A small preview must not silently redefine an export, and an uncovered date must not become a zero-count day. [\[1\]](#) [\[2\]](#)

Implemented contribution

The core contribution is a provider-neutral analytical boundary. The model need not implement arithmetic, invent query semantics or carry an entire result table in context. Version 0.6 adds bounded normalization, exact export admission and requested-field retrieval, grounded in a static scale review rather than a newly executed scale experiment. [\[1\]](#) [\[6\]](#)

Evidence class: synthetic fixture analyses, generated local workload observations and focused deterministic tests. One billion records is a design target, not a measured capacity result. [\[1\]](#)

1. Research framing and semantics

Text-to-query research distinguishes understanding a question from executing its program. Spider evaluates transfer to new databases and complex query structures; it motivates keeping natural-language interpretation separate from deterministic backend correctness. This project targets a fixed civic catalog and Elasticsearch, so it is not a Spider reproduction or cross-domain accuracy claim. [8]

Elastic documents search_after with point-in-time contexts to preserve a stable view across deep pagination. That guidance motivates the adapter's snapshot and cursor semantics. It does not establish that the local implementation has passed a live-engine performance evaluation. [9] [10]

Formal analytical object

Let D_v be the immutable dataset version and $q=(v,F,I,G,M,Z)$ a validated specification: version, non-time filter, half-open interval, grouping, metric definitions and timezone. Let $C(q)=\{r \text{ in } D_v: F(r) \text{ and time}(r) \text{ in } I\}$. Preview, CSV and map are projections of this saved cohort, not independent reconstructions from the user's natural language. [2]

```
n(g,I) = number of cohort records in group g
rate(g,I) = n(g,I) / calendar_exposure_days(I,Z)
change(g) = rate(g,I_after) - rate(g,I_before)
mean_closure(g) = sum(valid durations) / count(valid durations)
```

Mean closure describes administrative elapsed time from creation to closure, not first response or verified resolution. Its denominator is neither all requests nor all closed requests when usable timestamps are missing. Calendar exposure uses the requested timezone; elapsed hours are a different quantity. The cohort describes reported requests, not incident incidence or causal agency performance. [2] [3]

Prospective research question: can a bounded analytical contract improve end-to-end semantic correctness and artifact agreement across different agents without unacceptable interaction cost? Current fixture outcomes establish selected behaviors, not a population-level answer.

2. Method and implemented architecture

A request selects records, aggregate or compare_periods and binds that operation to a dataset version. Filters can express bounded Boolean predicates, category families, geographic regions and explicit time intervals. Unsupported combinations return a typed error instead of a silently simplified query. [2]

Phase	Service responsibility
Discover	Return the catalog, field meanings, coverage, limits and supported request examples.
Validate	Resolve absolute time boundaries, check schema and reject unsupported or uncovered analyses.
Execute	Compute the selected records or aggregates and save normalized specification, dataset identity and execution flags.
Inspect	Return bounded pages with exact matching totals and explicit truncation or approximation.
Publish artifact	Derive CSV or map from the saved cohort; retain scope, excluded geography and artifact hashes.

Time, rates and missingness

Time intervals are start-inclusive and end-exclusive. Relative presets resolve through a shared timezone library and are saved as absolute boundaries. Period comparisons use the same non-time filters and grouping domain; they report counts, exposure days, daily rates, absolute differences and nullable relative change. Complete creation-date coverage is required for both periods. [2]

Closure duration measures administrative created-to-closed time, not time to first response or verified resolution. Missing usable durations remain null and are excluded from duration statistics while closed-request counts retain those records. A null value therefore cannot be interpreted as zero hours. These definitions travel with results instead of depending on the agent to recall them. [2] [3]

Artifacts retain analytical scope

CSV and map tools consume a saved result ID. They require an explicit all-matching or selected-groups scope. A records export reproduces the full selected cohort rather than the preview. Version 0.6 checks exact planned rows before reserving a worker and limits backend source retrieval to selected CSV fields while retaining selection and ordering semantics. [2] [6]

3. Execution and artifact invariants

The service records the normalized specification, catalog and dataset identity together with its result. This supports a useful invariant: a later artifact must refer to the same analytical selection, even if the initial result was truncated for display. The following pseudocode summarizes that contract. [2]

```
discover catalog and immutable dataset identity
validate specification, operation, coverage and limits
execute compiled query; save result envelope
load saved result; verify version and explicit scope
CSV branch:
check exact row admission; reserve background worker
stream stable rows; verify counts; publish CSV + manifest
Maps branch:
verify exact-index data view without an implicit time field
points: build saved DSL filter; count located / missing rows
trends: publish and verify bounded saved NTA metrics
request Kibana locator link for configured map / join layer
```

Failure is a typed outcome

Invalid requests, insufficient coverage, unsupported operations, exceeded budgets and partial execution are distinct errors. An empty complete result is valid evidence of no matching records; a timeout or unobserved date is not. CSV status counters describe persisted progress, not worker liveness. Maps processing does not use the CSV worker: a returned link still requires rendered filter, value and boundary-join verification. [2] [11]

Why admission and streaming both matter

Exact row admission prevents a known oversized task from consuming a worker slot, but byte size still depends on encoded fields. Stream-time limits are therefore necessary after admission. Disk observations can prevent obvious exhaustion without reserving the space; cooperative deadlines can overrun a blocked I/O call. These distinctions shape how the interface describes pending and failed artifacts. [6] [11]

For N selected rows and k projected columns, serialization work is proportional to encoded output bytes, not merely $N \times k$. Backend aggregation costs depend on cardinality and index layout, while agent latency includes tool planning and synthesis. Separating these costs avoids extrapolating a small fixture benchmark into physical cluster capacity.

4. Worked example

The authored demonstration interprets Brooklyn noise last month relative to January 2, 2026. The saved window is December 2025 in America/New_York. It finds four matching synthetic requests, previews two and exports all four. The example shows why preview size and cohort size must be different fields. [3]

Question	Observed fixture result	Meaning
December Brooklyn noise	4 requests; 2-row preview; 4-row CSV	Export preserves full selection.
November-December rodent comparison	17 requests across 6 groups	Rates use 30 versus 31 days.
Brooklyn DOHMH closure	7 requests; 6 closed; mean 31.2 h	Mean uses 5 usable durations.
Brooklyn NYPD closure	7 requests; 5 closed; mean 3.6 h	Different request mixes prevent causal comparison.

Why unchanged counts can have changed rates

A fixture neighborhood with one request in each month has November rate 1/30 and December rate 1/31 requests per day. December minus November is -0.0010752688 requests per day. The count is unchanged; exposure differs. This arithmetic is descriptive reported-request activity, not incident prevalence, statistical significance or a causal effect. [3]

Unknown is an analytical result

The same demonstration asks for January-February 2026 rodent trends, outside the fixture coverage. The service returns coverage_gap rather than filling unobserved dates with zeros. The behavior protects the meaning of a comparison before any model-generated narrative is written. [3]

Neighborhood labels in this demonstration are explicitly synthetic. Real neighborhood analyses require versioned official geometry and a defined assignment policy. A map of underlying complaint points is also analytically different from a neighborhood trend layer; the artifact contract preserves that distinction. [2] [3]

Table 1. All numerical results on this page describe the 32-record fixture. They are not findings about actual New York City residents or agencies.

5. Recorded results and interpretation

Evidence	Recorded result	Scope
Version 0.6 focused checks	119 passed; no failures, errors or skips	Selected deterministic checks; tiny authored records and fake transports. [5]
Generated local workload	100,000 rows; 3 runs	Fixture backend, 15 returned groups per run. [4]
Historical run times	13.0741, 4.5881, 15.2276 s	Median 13.0741 s; cache states uncontrolled. [4]
Current scale review	Static reasoning and implementation	No scale simulation or live engine run for v0.6. [6]

The historical generated workload exercised tool execution, not model latency or natural-language accuracy. With only three runs and uncontrolled caches, its maximum is an observed maximum rather than a stable tail-latency estimate. The fixture backend cannot establish Elasticsearch or Kibana capacity. [4]

Resource-aware implementation

Version 0.6 limits normalization output bytes, checks observed free disk, uses portable UTF-8/LF output and records limits in the manifest. Known oversized CSV cohorts are rejected before worker admission. Streaming row, byte, deadline and source-identity guards remain active after admission: a planned row count is not a promise of successful completion. [2] [6]

Interpretation

These results support a narrower but useful conclusion: analytical semantics can be implemented independently of an agent and tested through inspectable inputs and outputs. The worked example verifies distinctions that conversational systems often blur, including missing durations, full cohorts, exposure-normalized rates and date coverage. It does not estimate how often an arbitrary agent chooses the correct specification.

The service defines a path from small local fixtures to self-hosted indexing. Its capacity claims remain tied to the measured backend and workload. Real-city analytical validity requires a qualified immutable dataset and independent reference answers; the synthetic numbers presented here are deliberately interpreted only within their authored environment. [1] [3]

6. Proposed end-to-end evaluation

PROPOSED STUDY. The real-data and agent evaluation below is a research plan, not a claim that the million-record or live visualization gates have been passed.

Layer	Independent reference	Evaluation unit
Dataset integrity	Unique keys, frozen revisions, date coverage and geometry versions	Qualified immutable capture
Deterministic analytics	Separately calculated counts, rates and valid-duration statistics	Query and complete result cohort
Agent interpretation	Human-authored questions with accepted/clarify specifications	Question across repeated runs and agent hosts
Artifacts	CSV key set and map cohort/metric comparison	Saved result, export and rendered map
Resources	Measured hardware, index size, latency and failure recovery	Workload profile and concurrent task count

The project acceptance design specifies at least two million real unique requests and a held-out set of 40 questions repeated three times, with a second agent comparison. These are planned gates, not achieved measurements. Query families should include dates, multiple filters, geography, grouping, comparisons and clarification cases, with final prompts hidden from implementation tuning. [1] [10]

Score intent correctness separately from execution correctness. Count a question as artifact-consistent only when narrative numbers, saved result, full CSV selection and requested map metric agree. For geography, report both matched and unmapped counts. Preserve all attempted runs, errors and unsupported questions in denominators; selective success-only timing would bias the result.

Use paired question-level comparisons between agent hosts and report uncertainty at the question level, with repeated outputs nested within questions. Resource experiments should vary row count, filter selectivity, group cardinality and export width independently. Measure actual cold/warm conditions and sustained concurrency rather than deriving tail latency from three observations.

7. Roadmap and analytical validity

Phase	Research artifact	Progression criterion
Qualify data	Frozen real capture, uniqueness audit and versioned geography	Coverage claims independently justified.
Verify parity	Live backend, full CSV and rendered Maps comparisons	Agreement across declared query families.
Evaluate agents	Held-out question set, repeated runs and second host	Intent/answer/artifact scores reported together.
Measure operations	Hardware-specific scale and recovery report	Capacity claims tied to observed workloads.

The present implementation is In progress. The roadmap strengthens evidence for an existing service and does not project performance numbers or a completion date. Static reasoning can improve admission and storage safeguards before expensive workloads, but only execution on a qualified target can support a measured capacity claim. [6] [10]

Threats to validity

Data capture is a source of uncertainty independent of query correctness. A provider's count and sampled records may be internally consistent without proving a transactionally complete local population. Complaint rates reflect reporting behavior and access as well as underlying conditions. Administrative closure is a process field, not a direct observation of first response or resolution. [3] [10]

Agent demonstrations may benefit from examples exposed by tool discovery, while development fixtures make difficult cases unusually legible. Final questions should test unseen combinations and realistic ambiguity. Engine metrics that approximate percentiles must retain that approximation rather than borrowing the fixture engine's exactness. [2] [3]

Conclusion

The implemented service makes the analytical specification and selected population durable objects. The recorded fixture demonstrates how this protects rates, missingness, coverage and full exports. Its research value lies in a reproducible architecture and falsifiable evaluation plan; broad conversational and physical-scale claims remain tied to future measured evidence.

8. Reproducibility and research record

The README and runtime guide describe installation, dataset configuration, ingestion and MCP connection. Begin with the local fixture before selecting an external backend. Use `describe_dataset` to obtain the current catalog and request rules; save the returned result identity before deriving artifacts. [1] [7]

```
python -m pip install -e "[mcp,geo]"
python -m analytics311 doctor
python -m analytics311 run examples/brooklyn-noise.json
python -m analytics311 mcp
```

Commands run from the nested project directory in an appropriate Python environment. The bundled fixture operates locally after dependency installation. The original authored demonstration includes its CLI trace, request specifications and exported CSV fingerprint. The v0.6 focused receipt identifies its source/test selection; it should be read as focused verification for that version. [1] [3] [5]

The PDF summarizes retained source records; publication did not rerun the historical benchmark or launch a live Elasticsearch workload. Refer to the evidence manifest for source hashes.

A reproducible research record

The public nested project source contains implementation, documented contracts and retained development evidence. The publication evidence manifest records SHA-256 fingerprints for each local source cited in this paper. External references link to primary publications or official technical documentation, verified for this edition.

A future rerun should record the exact commit, dependency/runtime identities, input hashes, configuration and environment, plus each attempt and its outcome. Keep generated outputs separate from source and mark any changed implementation as a new condition. Reproduction of a software check does not automatically reproduce a model judgment, benchmark distribution or independent human label.

Authorship and project status

Boomer Rawlings developed this work individually after the original submission and grading dates. The current project is In progress. The implemented methods and recorded observations in this paper are separated from the explicitly proposed research protocol and roadmap. No group-project, graded-submission or completed-research claim is made.

References

Repository references identify the evidence used for the implementation and result claims. Primary literature supports the research framing; its published results are not attributed to this project.

[1] [Implemented service and current version](#)
IMPLEMENTATION.md

[2] [Analysis and artifact tool contract](#)
docs/tool-contract.md

[3] [Authored CLI demonstration results](#)
examples/agent-demo/answer.md

[4] [Generated 100,000-record measurements](#)
examples/evidence/generated-100000-benchmark.json

[5] [Version 0.6 focused verification receipt](#)
examples/evidence/test-report-v0.6-focused.json

[6] [Static scale review and version 0.6 changes](#)
docs/forecast-v0.6.md

[7] [Local runtime and ingestion instructions](#)
docs/runtime.md

[8] Yu et al. (2018). Spider: A Large-Scale Human-Labeled Dataset for Complex and Cross-Domain Semantic Parsing and Text-to-SQL Task. EMNLP.
<https://arxiv.org/abs/1809.08887>

[9] Elastic. Paginate search results. Elasticsearch reference, accessed October 7, 2026.
<https://www.elastic.co/docs/reference/elasticsearch/rest-apis/paginate-search-results>

[10] [Project verification and live-evaluation requirements](#)
docs/acceptance.md

[11] [Reproducible export and ingestion recovery](#)
docs/reliability.md